

YIJUN BIAN

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🔗 <https://scholar.google.com/citations?user=SmvfHHgAAAAJ>

🏠 <https://eustomadew.github.io>

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Employment

01/08/2024 to present	MSCA Postdoc , University of Copenhagen (UCPH) Machine Learning Section, Dept. Computer Science Marie Skłodowska-Curie Actions (MSCA) Postdoctoral Fellowship	Copenhagen, Denmark <i>Supervisor: Prof. Yevgeny Seldin</i>
24/03/2023 – 30/07/2024	Research fellow , National University of Singapore	Singapore
02/12/2020 – 22/03/2023	Algorithm engineer , Shanghai Institute of Microsystem & Information Technology	Shanghai, China
01/11/2018 – 30/04/2019	Visiting research scholar , Texas A&M University DATA Lab, Dept. Computer Science & Engineering	College Station, TX, United States <i>Adviser: Prof. Xia (Ben) Hu</i>
01/03/2016 – 31/07/2016	Teaching assistant , USTC Course: Mathematical Analysis (B2), School of Mathematical Sciences	Hefei, Anhui, China
02/09/2014 – 20/11/2020	Graduate research assistant , USTC Topics: machine learning, ensemble learning Dept. Computer Science & Technology, University of Science and Technology of China	Hefei, Anhui, China <i>Supervisor: Prof. Huanhuan Chen</i>

Education

02/09/2014 – 20/11/2020	University of Science and Technology of China (USTC)	Hefei, Anhui, China
07/12/2020	Awarded <i>Doctor of Engineering</i>	<i>Supervisor: Prof. Huanhuan Chen</i>
26/08/2020	Dissertation defence: Research and Applications of Diversity in Ensemble Classification Ph.D. in Computer Science & Technology, successive postgraduate and doctoral programme Overall GPA: 3.65/4.30 (Grade: 86.87/100.00) Ranking: 2/37 (10/114 as MSc student)	
02/09/2010 – 01/07/2014	Northwest A&F University (NWAU)	Yangling, Shaanxi, China
01/07/2014	Awarded <i>Bachelor of Natural Sciences</i> , College of Science B.Sc. in Information and Computing Science (aka. Computational Mathematics) Overall GPA: 3.44/4.00 (Grade: 87.68/100.00) Ranking: 5/47	

Publications¹

Peer-reviewed articles

- A4 Ming Chen, **Yijun Bian**, Nanguang Chen, and Anqi Qiu*, “Orthogonal mixed-effects modeling for high-dimensional longitudinal data: An unsupervised learning approach,” in *IEEE Transactions on Medical Imaging*, vol. 44, no. 1, pp. 207–220, Jan 2025, doi: [10.1109/TMI.2024.3435855](https://doi.org/10.1109/TMI.2024.3435855).
- A3 **Yijun Bian**, Qingquan Song, Mengnan Du, Jun Yao, Huanhuan Chen*, and Xia Hu, “Subarchitecture ensemble pruning in neural architecture search,” in *IEEE Transactions on Neural Networks and Learning Systems*, vol. 33, no. 12, pp. 7928–7936, Dec 2022, doi: [10.1109/TNNLS.2021.3085299](https://doi.org/10.1109/TNNLS.2021.3085299).
- A2 **Yijun Bian** and Huanhuan Chen*, “When does diversity help generalization in classification ensembles?” in *IEEE Transactions on Cybernetics*, vol. 52, no. 9, pp. 9059–9075, Sept 2022, doi: [10.1109/TCYB.2021.3053165](https://doi.org/10.1109/TCYB.2021.3053165).
- A1 **Yijun Bian**, Yijun Wang, Yaqiang Yao, and Huanhuan Chen*, “Ensemble pruning based on objection maximization with a general distributed framework,” in *IEEE Transactions on Neural Networks and Learning Systems*, vol. 31, no. 9, pp. 3766–3774, Sept 2020, doi: [10.1109/TNNLS.2019.2945116](https://doi.org/10.1109/TNNLS.2019.2945116).

Peer-reviewed conference paper

- C1 Abhijith Sharma, **Yijun Bian**, Vatsal Nanda, Phil Munz, and Apurva Narayan*, “Vulnerability of CNNs against multi-patch attacks,” in *Proceedings of the 2023 ACM Workshop on Secure and Trustworthy Cyber-Physical Systems (SaT-CPS’23)*, pp. 23–32, Apr 2023, doi: [10.1145/3579988.3585054](https://doi.org/10.1145/3579988.3585054).

Peer-reviewed non-archival posters

- P2 **Yijun Bian*** and Kun Zhang, “Increasing fairness via combination with learning guarantees,” in *NeurIPS 2024 Workshop on Mathematics of Modern Machine Learning (M3L)*, Dec 2024.
- P1 **Yijun Bian***, Yujie Luo#, and Ping Xu, “Does machine bring in extra bias in learning? Approximating discrimination within models quickly,” in *NeurIPS 2024 Workshop on Mathematics of Modern Machine Learning (M3L)*, Dec 2024.

Manuscripts under review

- U9 **Yijun Bian*** and Lei You, “Fairness research for machine learning should integrate societal considerations,” *arXiv preprint [arXiv:2506.12556](https://arxiv.org/abs/2506.12556)*, 2025.
- U8 Lin Zhu#, **Yijun Bian**#, and Lei You*, “FairSHAP: Preprocessing for fairness through attribution-based data augmentation,” *arXiv preprint [arXiv:2505.11111](https://arxiv.org/abs/2505.11111)*, 2025.
- U7 Lei You*, **Yijun Bian**, and Lele Cao, “Refining counterfactual explanations with joint-distribution-informed Shapley towards actionable minimality,” *arXiv preprint [arXiv:2410.05419](https://arxiv.org/abs/2410.05419)*, 2024.
- U5 **Yijun Bian***, Yujie Luo#, and Ping Xu, “Approximating discrimination within models when faced with several non-binary sensitive attributes,” *arXiv preprint [arXiv:2408.06099](https://arxiv.org/abs/2408.06099)*, 2024.

- U3 Jinghan Huang[#], Qiufeng Chen[#], **Yijun Bian**, Pengli Zhu, Nanguang Chen, Moo K. Chung, and Anqi Qiu*, “Advancing graph neural networks with HL-HGAT: A Hodge-Laplacian and attention mechanism approach for heterogeneous graph-structured data,” *arXiv preprint* [arXiv:2403.06687](https://arxiv.org/abs/2403.06687), 2023.
- U2 **Yijun Bian*** and Kun Zhang, “Increasing fairness via combination with learning guarantees,” *arXiv preprint* [arXiv:2301.10813](https://arxiv.org/abs/2301.10813), 2023.

Manuscripts in revisions

- U6 Alina Basharat, **Yijun Bian**, Ping Xu*, and Zhi Tian, “Towards trustworthy federated learning,” *arXiv preprint* [arXiv:2503.03684](https://arxiv.org/abs/2503.03684), 2025.
- U4 **Yijun Bian**^{#*} and Yujie Luo[#], “Does machine bring in extra bias in learning? Approximating fairness in models promptly,” *arXiv preprint* [arXiv:2405.09251](https://arxiv.org/abs/2405.09251), 2024.
- U1 Abhijith Sharma[#], **Yijun Bian**^{#*}, Phil Munz, and Apurva Narayan, “Adversarial patch attacks and defences in vision-based tasks: A survey,” *arXiv preprint* [arXiv:2206.08304](https://arxiv.org/abs/2206.08304), 2022.

Released code of research projects

- 17/01/2024 – 05/09/2024 **ApproxBias** (URL: <https://github.com/eustomaqua/ApproxBias>). Official released code for the papers U4&U5, that is, “Does Machine Bring in Extra Bias in Learning? Approximating Fairness in Models Promptly” and “Approximating Discrimination Within Models When Faced With Several Non-Binary Sensitive Attributes.”
- 16/08/2022 – 29/09/2022 **SAEP** (URL: <https://github.com/eustomaqua/SAEP>). Official released code for the paper entitled “Subarchitecture Ensemble Pruning in Neural Architecture Search.”
- 16/04/2020 – 14/09/2021 **EPFD** (URL: <https://github.com/eustomaqua/EPFD>). Official released code for the paper entitled “Ensemble Pruning Based on Objection Maximization With a General Distributed Framework.”
- 24/07/2019 – 19/04/2020 **PyEnsemble** (URL: <https://github.com/eustomaqua/PyEnsemble>). This is an open-source library for ensemble learning methods, diversity measures, and ensemble pruning methods.

Teaching and supervision

- 30/05/2025 to present **Supervision to two MSc students** Copenhagen, Denmark
Supervising two MSc students, as part of the "Project Outside the Course Scope" program
- 27/01/2025 to present **Supervision to two BSc students** Copenhagen, Denmark
Supervising two BSc students with their Bachelor thesis projects
- 27/10/2024 – 20/05/2025 **Co-supervision to one MSc student** Virtually
Helped supervise one MSc student with research and scientific manuscripts, in collaboration with Prof. You at the Technical University of Denmark
- 31/07/2023 – 24/07/2024 **Co-supervision to two MEng students** Singapore
Helped supervise two MEng students with research and scientific manuscripts, in collaboration with Prof. Qiu at the National University of Singapore
- 06/03/2022 – 03/02/2023 **Co-supervision to one MSc student** Remotely
Helped supervise a MSc student with research and scientific manuscripts, in online collaboration with Prof. Narayan at the University of British Columbia
- 01/03/2016 – 31/07/2016 **Teaching assistant, USTC** Hefei, Anhui, China
Course: Mathematical Analysis (B2), School of Mathematical Sciences
Duties included grading the homework and examination papers of students, taking exercise classes, and answering questions from students

Other scientific qualifications

Journal reviewer

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|-----------------------|--|-----------------|
| 13/03/2025 to present | IEEE Transactions on Image Processing | one completed |
| 28/05/2024 to present | Information Sciences | two completed |
| 13/01/2024 to present | IEEE Transactions on Knowledge and Data Engineering | one completed |
| 06/06/2023 to present | Scientific Reports | three completed |
| 31/05/2023 to present | IEEE Transactions on Pattern Analysis and Machine Intelligence | five completed |
| 05/03/2023 to present | Journal of Supercomputing | one completed |
| 23/09/2021 to present | IEEE Transactions on Emerging Topics in Computational Intelligence | seven completed |
| 13/11/2020 to present | Neural Networks | eight completed |
| 06/08/2016 to present | IEEE Transactions on Neural Networks and Learning Systems | 19 completed |

Conference reviewer

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| 31/05/2025 to present | AIES'25 (AAAI/ACM Conference on AI, Ethics, and Society) | three ongoing |
| 03/10/2024 – 08/10/2024 | M3L'24 (NeurIPS Workshop, Mathematics of Modern Machine Learning) | one completed |
| 30/11/2023 – 15/12/2023 | PPAI'24 (AAAI Workshop on Privacy-Preserving Artificial Intelligence) | two completed |

Selected honours and awards

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| 15/11/2023 | Grant awardee, under the Horizon Europe Marie Skłodowska-Curie Actions Postdoctoral Fellowships (MSCA PF) call 2022 | European Commission |
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04/04/2023	Seal of Excellence certificate, under the MSCA PF call 2022	European Commission
01/11/2019	GDC (Awarded from the Global Digital Creations Holdings Limited) scholarship	USTC
19/04/2018	International Exchange Funding for Excellent Students	USTC
11/09/2018, 30/09/2017, 30/09/2016	Second-class academic scholarship, for PhD students	USTC
29/09/2015	First-class academic scholarship, for MSc students	USTC
30/09/2014	Second-class academic scholarship, for MSc students	USTC
30/06/2014	Outstanding undergraduate graduation thesis (design)	NWAFU
20/12/2013	President scholarship	NWAFU
31/01/2013	Excellence award of the undergraduate innovation forum and finding presentation	NWAFU
31/12/2013, 06/12/2012, 20/12/2011	Merit student, for three consecutive years	NWAFU
31/10/2013, 31/10/2012, 31/10/2011, 31/03/2011	First-class professional scholarship, four times in a row	NWAFU

Selected outreach activities

Invited presentations

01/05/2025	“Fairness evaluation in ML. Why would we care?” Guest Lecture for 62533/62T22 ("Applied Machine Learning and Big Data", Technical University of Denmark)	DTU
05/12/2024	“Study of FAIR ML models from a theoretical perspective,” DIKU MIA Seminar (Medical Image Analysis, Department of Computer Science)	UCPH
26/10/2023	“Increasing fairness via combination with learning guarantees,” Fall 2023 LeT-All mentorship workshop (Session 2, Practice talk), Learning Theory Alliance (LeT-All)	Virtually

Open source contributor

2020	Arctic Code Vault Contributor in the 2020 GitHub Archive Program
15/10/2019 – 17/10/2019	<i>AdaNet</i> (Google’s open-source project), merged pull request
12/08/2019	<i>OpenNE</i> , merged pull request
10/06/2019 – 13/06/2019	<i>AutoKeras</i> , merged pull requests a and b in the blocks branch

¹#Equal contribution; *Corresponding author.